



Neural network modeling of in situ fluid-filled pore size distributions in subsurface shale reservoirs under data constraints

Hao Li¹ · Siddharth Misra¹ · Jiabo He¹

Received: 21 February 2018 / Accepted: 26 February 2019 / Published online: 8 March 2019
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Abstract

Subsurface nuclear magnetic resonance (NMR) logs acquired in the wellbore environment are sensitive to fluid-filled pore size distribution, fluid mobility, permeability, and porosity in the near-wellbore reservoir volume. NMR response of a formation layer is processed to extract the T2 distribution, which approximates the fluid-filled pore size distribution. NMR logs are acquired in limited number of wells due to financial and operational challenges, which adversely affects reservoir characterization. We developed two neural-network-based machine learning techniques, long short-term memory (LSTM) network and variational autoencoder with a convolutional layer (VAEc) network, to process the ‘easy-to-acquire’ formation mineral and fluid saturation logs to generate synthetic NMR T2 distributions in the absence of ‘hard-to-acquire’ NMR T2 distribution log. Both the predictive models are trained and tested on limited wireline log measurements randomly selected from a 300-ft depth interval of the Bakken shale formation. Synthesis performances of LSTM and VAEc models in terms of R^2 are 0.78 and 0.75, respectively. Noise is inevitable in logging data due to the complex wellbore and formation conditions. Notably, both the predictive models robustly synthesize the fluid-filled pore size distributions in the presence of 50% noise in input logs and 30% noise in training T2 data. The performance of the proposed methodology improves with access to larger volume of training data from other formation types. The proposed method is critical to the synthesis of in situ fluid-filled pore size distributions in shale formations under data constraints due to financial and operational challenges.

Keywords Convolution · Recurrent · Neural network · Shale · Nuclear magnetic resonance · Deep learning

1 Introduction

A well log is a measurement of certain physical property of subsurface geological formation located at a depth of several thousand feet below the ground. Well logs are acquired through wellbore, 4 to 6 inches in diameter, drilled into these formations to access them for purposes of energy and earth resource extraction. The well logs are

interpreted using empirical and mechanistic models for near-wellbore reservoir characterization. Well logging tools are run in the wellbore during drilling or after drilling in an open-hole or a cased-hole environment. The borehole and formation conditions in which the well logging tools operate are uncertain, complex, and hostile that give rise to noise-prone measurements.

One advanced well log is NMR measurement, which is sensitive to hydrocarbon and water volumes present in the connected pore volumes. Subsurface NMR log is valuable for reservoir characterization because it contains physical information about the pore-filling fluid distribution, type, and mobility. Porosity and permeability of a reservoir are routinely estimated from NMR-T2 distribution, computed from the NMR log. However, the acquisition of NMR log is limited due to the high cost and operational challenge in deploying and running these tools in the uncertain, complex, and hostile borehole environment. In shale

✉ Siddharth Misra
misra@ou.edu

Hao Li
haoli@ou.edu

Jiabo He
jiabo.he@ou.edu

¹ Mewbourne School of Petroleum and Geological Engineering, The University of Oklahoma, Norman, OK 73069, USA

formations, NMR log acquisition is not common due to operational and financial challenges. In such data-constrained environment, new methods are required to determine the fluid-filled pore size distribution in hydrocarbon-bearing shale reservoirs.

Machine learning (ML) aims to extract patterns and hidden relationships in data. ML technique formulates a data-driven predictive model, which is useful when the patterns and relationships among data entities are governed by complex processes and mechanisms. Applications of CNN are common for image processing, which may involve object detection, face recognition, and image captioning. RNN is assisted by an internal memory to processes a sequential data. Long short-term memory (LSTM) is a popular type of RNN due to its ability to ‘memorize’ relatively longer-term dependencies in data. LSTM has been applied to speech recognition, language translation, and image captioning [1]. Basic artificial neural networks (ANN) and shallow predictive models have been applied in oil production prediction [2–4], pressure transient analysis [5, 6], and history matching [7]. However, the advanced machine learning methods, involving RNN and CNN, have not yet been widely applied in the petroleum industry. In this paper, two neural-network-based machine learning techniques, LSTM network and variational autoencoder with a convolutional layer (VAEc) network, process easy-to-acquire formation mineral and fluid saturation logs to generate the NMR T₂ distributions, which approximate fluid-filled pore size distributions in the Bakken shale (BS) formation.

A few studies have been conducted to implement advanced ML methods in well-log-based reservoir characterization. Mohaghegh et al. [8] used a five-layer ANN to generate synthetic irreducible water saturation, effective porosity and permeability from gamma ray, caliper, resistivity, and spontaneous-potential logs from 4 different wells. Jamshidian et al. [9] used ANN model to determine the NMR-derived free-fluid porosity and permeability from conventional well logs of sonic, neutron porosity, bulk density, and resistivity from the South Pars gas field. Labani et al. [10] estimated NMR-derived free-fluid porosity and permeability from conventional logs, such as sonic, resistivity, and gamma ray. Instead of predicting reservoir properties, a few studies predicted well log by processing conventional logs. Golsanami et al. [11] combined models of fuzzy logic and ANN to predict the eight-bin porosities of NMR T₂ distribution from neutron, density, sonic, lithology, gamma ray, and effective porosity logs acquired from a reservoir unit consisting of limestone, dolomite, and shale. Farzi et al. [12] simulated NMR response from micro-CT images of six sand packs and two sandstone samples using an ANN. Asoodeh and Bagheripour [13] applied a committee machine to generate

compressional, shear, and Stoneley wave velocities from conventional log data. Recent studies show promising results about applying deep learning models for NMR T₂ spectral generations [14, 15].

In this study, we implement variational autoencoder with a convolutional layer (VAEc) network as the first model. CNN is inspired by the complex arrangement of the biological visual cortex, where a set of neurons is sensitive to a specific visual feature, such as angle, curvature, or edge. A convolutional layer in CNN contains a combination of mathematical filters, in which each filter is sensitive to a specific feature of an image. When multiple convolutional layers are combined, the CNN can learn features of varying visual complexity. Variational autoencoder is a special type of generative neural network that aims to reproduce the input of the network. It comprises two computational frameworks, an encoder followed by a decoder. The encoder encodes the input into a latent layer, whereas the decoder decodes the information in the latent layer to reconstruct the input. With this architecture, the VAE learns the data distribution in the input and generates the input data that are similar to the training data, and can project a high-dimensional data to a lower-dimensional space. As a generative model, the VAE has been applied in various generative modeling problems like image generation [16] and text generation [17]. We implement long short-term memory (LSTM) network as the second model. LSTM is a type of RNN with an improved ability to solve the gradient vanishing problem in RNN. With the ability of ‘memorizing’ values for a long time, the LSTM is superior in dealing with sequences, like text, where values in a different position may have a strong dependency. LSTM applications include machine translation [18, 19], natural language generation [20], and time series prediction. Joint application of LSTM and CNN can accomplish the work of video description [21], image captioning [1], and text to image generation [22].

Well logs have complex hidden relationships among each other. Although different well logs sense different physical phenomena and formation properties, certain combinations of well logs tend to correlate because they are controlled by certain combinations of formation properties and physical processes, such as mineral composition, fluid type, fluid saturations, pore size distribution, fracture distribution, to name a few. Under data-constrained environments due to financial or operational challenges, the goal of this study is to utilize LSTM and VAEc models to extract the complex hidden relationships between the easy-to-acquire well logs and the NMR T₂ distribution logs, which approximate the fluid-filled pore size distribution. In doing so, fluid-filled pore size distribution in Bakken shale formation can be synthesized by processing 10 easy-to-acquire formation mineral and fluid saturation logs.

The paper is organized as follows: Sect. 2 explains the theory NMR measurements, LSTM, VAE, and CNN; Sect. 3 explains the data preprocessing; Sect. 4 presents the results of data-driven synthesis of NMR; and Sect. 5 presents the performance and robustness of the models.

2 Theory and methods

2.1 NMR measurement theories

NMR measurement utilizes the polarization and relaxation phenomena of hydrogen nuclei in the directional magnetic field to investigate the fluid properties and fluid distributions in porous materials. In oil and gas reservoirs, hydrocarbon and water contain the hydrogen nuclei that respond to the external magnetic excitation. Therefore, subsurface NMR logs measure the mobility and distribution of reservoir fluid in the subsurface formations. Hydrogen nuclei in the formation fluids are first polarized by a strong magnetic field exerted by the NMR logging tool in the borehole. Subsequently, the magnetic field is removed resulting in the polarized nuclei to lose energy and relax to an equilibrium state. This transition emits magnetic signals and is detected and recorded as NMR raw logs. NMR raw logs can be fitted by a set of exponentials with different decay constants [23]. The decay constants in these exponentials are the T2 times, which is a function of the bulk fluid relaxation ($T_{2\text{bulk}}$), surface relaxation ($T_{2\text{surface}}$), and diffusion relaxation ($T_{2\text{diffusion}}$) mechanisms. The relationship can be expressed as:

$$\frac{1}{T_2} = \frac{1}{T_{2\text{bulk}}} + \frac{1}{T_{2\text{surface}}} + \frac{1}{T_{2\text{diffusion}}} \quad (1)$$

Bulk relaxation is controlled by volumetric properties, such as pore volume, fluid saturation, fluid mobility, and fluid composition. Surface relaxation occurs at the fluid–solid interface when hydrogen nuclei interact with the grain surface, it is influenced by grain surface area, grain surface to pore volume ratio, grain mineralogy, and wettability [23]. Raw NMR logging data are processed to derive the NMR-T2 distribution at each depth along the borehole. NMR-T2 distribution is the relationship between NMR signal amplitude and T2 relaxation time. The T2 distribution represents near-wellbore pore-filling fluid distribution and mobility. T2 distribution is a 1D spectral response as a function of T2 time, where larger T2 time is correlated with larger pore volumes and larger fluid mobility. NMR log is the only log that can provide reservoir engineers with pore size distribution information of the reservoir. Pore size distribution describes the distribution of the pore diameter. Pore size distribution is critical in reservoir characterization. Pore size distribution can be used to calculate the

formation permeability, porosity, and rock typing essential to the reservoir characterization procedure.

2.2 Petrophysical basis for the proposed log generation

NMR logging tool is usually run at relatively slower speed of 2000 ft/h in bigger diameter uniform boreholes around 6.5-in in diameter. NMR signals due to the inherent physics have poor signal-to-noise ratio in the complex borehole conditions. Acquisition of high-quality NMR log in subsurface borehole environment requires highly trained wireline field engineers. Oil and gas companies do not deploy NMR logging tool in every well in a reservoir due to the financial and operational challenges involved in running the NMR tool. One alternative is to synthesize the entire NMR-T2 distribution spanning 0.3 ms to 3000 ms by processing conventional logs. An accurate prediction of NMR T2 distribution will assist the geoscientists and engineers to quantify the pore size distribution, permeability, and bound fluid saturation in hydrocarbon-bearing reservoirs, thereby improving project economics and reservoir characterization capabilities. Synthesis of the entire NMR T2 without core data is a challenging and novel task.

Generation of NMR T2 distribution log using conventional “easy-to-acquire” logs is feasible because these logs are sensitive to fluid saturations, pore size distribution, and mineralogy. Resistivity measurement and the mud-filtrate invasion effect on resistivity logs are influenced by the pore sizes, pore throat sizes, tortuosity, and Maxwell–Wagner polarization, which depends on the pore size distribution. Neutron log is sensitive to hydrogen index of fluids in the formation, which correlates with average porosity of the formation. GR is sensitive to clay volume that determines the clay-bound water and capillary-bound water, which is sensed by the NMR T2 distribution. Density log is sensitive to mineral densities and fluid densities in the formation. Mineral density is controlled by mineralogy that controls the lithology and pore size distribution of the formation. For example, sandstone formation with predominantly quartz mineralogy will have a distinct pore size distribution compared to carbonate formation with predominantly calcite mineralogy, which will be distinct from shale formation with predominantly clay mineralogy. NMR T2 distribution depends on fluid saturations, fluid mobility, pore size distribution, and surface relaxation, which depends on the mineralogy of the formation. These physical properties that govern NMR T2 also influence the logs, namely neutron, density, resistivity, sonic, and GR logs. The petrophysical basis for choosing these logs is because each of these logs has complex relationship with pore size distribution, fluid saturation, and mineralogy. We use

artificial neural networks to identify and extract these complex relationships that cannot be easily obtained by any other mechanistic models. The original goal of this study is not to develop a precise mechanistic equation to describe the physical relationships between the conventional logs and NMR T2 distribution. Rather, we train neural networks to learn these hidden patterns that relate conventional logs to the NMR T2 distribution. Another important concept that makes the NMR T2 generation possible is because most of these logs are influenced by water saturation and hydrocarbon saturation, and the vertical distribution of fluid saturations along the depth is controlled by capillary pressure, which is governed by the pore size distribution.

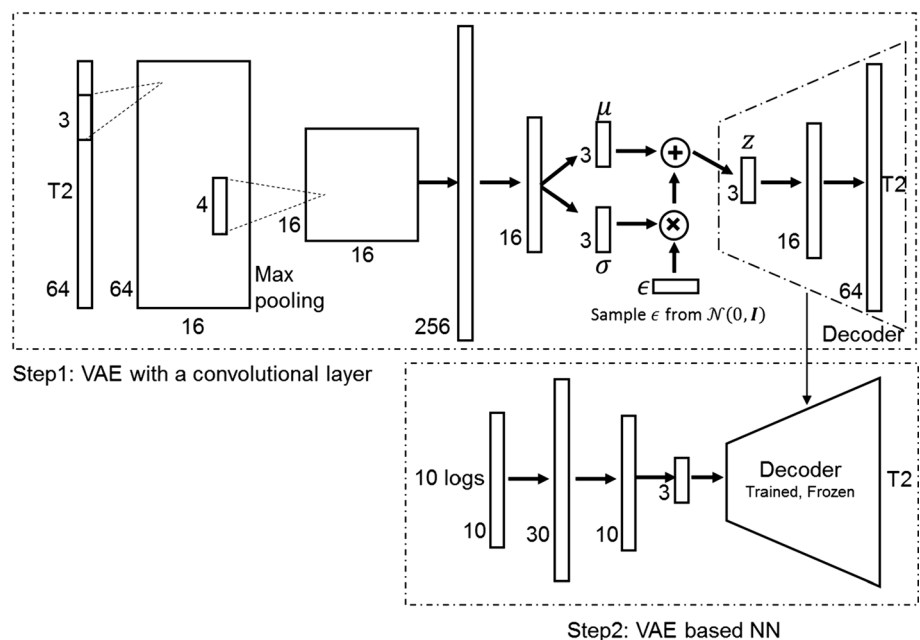
2.3 Architecture and training process of the VAEc model

A convolutional layer is combined with a VAE-based neural network (NN) to better extract features of NMR T2 distribution, which is processed as a 1D vector. The architecture of the generative model for purposes of VAEc-based NN model training is presented in Fig. 1. In the first part of training, VAEc learns to reproduce the measured T2 distributions. The encoder part of the VAE compresses the 64-dimension T2 data to 3-dimensional latent space. Subsequently, the decoder decompresses the latent variables to reconstruct the measured T2 input data. The first step of training, which involves the encoder and decoder, comprises 8 layers: 1 convolutional layer, 1 max pooling layer, 1 latent layer with 3 neurons, and 5 dense fully connected layers. The first convolutional layer is $64 \times 16 \times 1$ and is generated by filtering the 64×1 T2 distribution of a

specific depth with 16 filters of size 3×1 . The max pooling layer processes the $64 \times 16 \times 1$ output with a filter size of 4×1 to obtain a $16 \times 16 \times 1$ output. The output of the max pooling layer is then flattened and fed to two fully connected layers with 256 and 16 neurons each. The fifth layer of the VAEc is a 3-dimensional latent layer, which samples from the output of the previous layer. This layer is chosen to be 3 dimensional based on extensive numerical experiments. The 64-dimensional NMR T2 data are compressed to 3-dimensions through these five layers. After that, the decoder decodes the latent vector to 64-dimension through 3 fully connected layers containing 3, 16, and 64 neurons, respectively. The loss function in the VAEc model is comprised of two components: reconstruction loss and the Kullback–Leibler (KL) divergence loss. The reconstruction loss emphasizes the difference between the input and output NMR T2 data when forcing the VAEc to reproduce the input as precise as possible, whereas the KL divergence loss evaluates the difference between the distribution of the latent vector and the unit Gaussian distribution.

In the second step of training, a NN model comprising 4 layers is connected to the decoder trained in the first step. The trained decoder is frozen to preserve the generalization of the T2 distribution learned from the training data. The first three layers process 10 mineral composition and fluid saturation logs acquired at a specific depth in the subsurface to produce a 3-dimensional vector, which can be decoded by the trained decoder to generate the T2 distribution that best matches the T2 distribution measured at that depth. The architecture of the 4-layered NN model used in the second part of the training is shown Fig. 1.

Fig. 1 The architecture of the VAEc model during the training phase



2.4 Training process and architecture of LSTM

In the second predictive model, the 10 mineral composition and fluid saturation logs and NMR-T2 distribution measured in the subsurface along the length of a well are processed using LSTM as sequences. LSTM adopts a recurrent neural network architecture that remembers patterns over a sequence. LSTM is applied for sequence-to-sequence mapping, where a sequence is a list of ordered elements. LSTM excels other deep neural networks in that it can deal with long-term dependency and unfixed length of input data. This sequence-to-sequence learning model is successfully applied to language translation. Language translation relies on the fact that sentences in different languages are distinct representations of one common meaning. In a similar manner, different subsurface logs acquired at a specific depth are distinct responses to one common material. We implement LSTM to capture the dependency in NMR T2 distribution, which usually exhibits a smooth distribution. The smooth shape of NMR-T2 distribution illustrates the dependency between the data point in NMR T2.

The architecture of the LSTM model is demonstrated in Fig. 2. Like VAEc model, this model comprises 1 encoder followed by 1 decoder. Both encoder and decoder are collections of LSTM layers. The encoder and decoder LSTM layers are made up of 10 and 64 chained LSTM modules, respectively. Each module is made of gates controlling data flow inside the LSTM modules. Controlled by various gates, the LSTM can choose to forget or update the information flowing through the modules. The encoder compresses the 10 formation mineral and fluid saturation logs into a single vector. Then, the decoder decodes the vector into target sequences. The 10 input logs are taken as sequence, and the encoder of the model takes one of the ten logs at each time step. After all of the 10 input logs are read by the encoder, the encoder generates a vector v of size 15 in the last step. Size 15 is found based on experience with the LSTM predictions of T2 data. The vector v generated by the encoder is repeated 64 times and fed to the decoder.

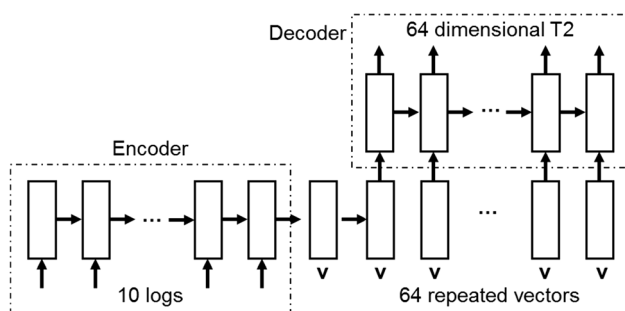


Fig. 2 The architecture of the LSTM model during training and testing

The decoder takes the 15-dimensional vector v in 64 time steps and tries to reconstruct a 64-dimensional T2 distribution for a single depth corresponding to the 10 input logs relevant to that depth.

The loss function used for the LSTM model is mean squared error function. The optimizer used to train the LSTM model is RMSprop. During the training process, the prediction loss is evaluated using the mean squared error function and RMSprop is used to update the weights during backpropagation.

2.5 VAEc-based NN model versus LSTM model

The convolutional layers in VAEc-based NN model tend to capture the features of the NMR T2 distribution better than the LSTM model. VAEc-based NN model is trained in a two-step process, which is more complicated than one-step training process of LSTM. During the VAEc model design process, we need to test different number of layers, different neurons, different number of filters, etc., in both training Step 1 and Step 2, whereas the LSTM model is trained in one step. Consequently, VAEc-based NN tends to require more time to tune the model parameters. Nonetheless, VAEc performs more accurate synthesis. In the Step 2 of the training, formation mineral and fluid compositions are processed using the trained decoder, which means the model is forced to synthesize NMR T2 distribution similar to those that were learnt in Step 1, thereby resulting in better accuracy. The decoder of LSTM is not pretrained during the training process, which means more effort is required to prevent overfitting through efficient architecture and training step selection. The LSTM model has an advantage of long sequence processing and can capture the dependency in the conventional logs as well as that in NMR T2. Usefulness of LSTM for the proposed objective will increase with an increase in the dependency in the log data. In our implementations, the LSTM model is deeper, whereas the VAEc-based NN model is wider compared to the other.

3 Data preprocessing

3.1 Bakken shale formation characteristics and data

The training and test data used in this study were derived from subsurface measurements acquired by various downhole logging tools deployed in 5- to 7-inch-diameter borehole drilled in the Bakken shale (BS) formation. The borehole intersected 7 distinct formation lithologies (Formation 1–7). Formations 1 and 3 are organic-rich hydrocarbon source shale rocks, and Formation 2 is a dolomite

reservoir. Total organic carbon (TOC) of Formations 1 and 3 ranges from 12 to 26 wt%, whereas the TOC of Formation 2 ranges from 0.1 to 0.3 wt%. Formations 4 to 7 are primarily dolomite and shale of poor to moderate reservoir quality.

Resistivity, neutron, density, natural gamma ray (GR), dielectric, and NMR logs were acquired in the Bakken shale formation. A few of these logs were inverted using Techlog mineral inversion software to obtain 10 formation mineral content and fluid saturation logs. The fluid-filled pore size distribution synthesis methodology proposed in this study is developed using the 10 inversion-derived formation mineral content and fluid saturation logs for training and testing of the NN models along with the NMR-T2 distributions for training the NN models.

3.2 Data preprocessing

Data preprocessing of the logs improves the efficiency of the proposed synthesis. Data cleaning and data normalization techniques were used in this study. Well log data are affected by environmental noise generated when the logging tool is run in the borehole and is also affected by the uncertainty due to physical variabilities in the surrounding volume. Thin beds and formation layers with unrepresented formation properties will decrease the training efficiency and synthesis accuracy. During data preprocessing, we remove the data corresponding to these thin layers that act as outliers. Data were acquired from 597 sequential discrete depths along the borehole intersecting the 300-ft depth interval of Bakken shale formation. Measured T2 distributions predominantly exhibit 1 or 2 peaks. Less than 4% of measured T2 distributions exhibit more than 2 peaks. Twenty-two data points that have 3-peak NMR T2 data were screened out from this study. The formation mineral content and fluid saturation logs are normalized to have zero mean and unit variance, which ensures the logs have the same scale and contribute equally to the loss function. After data cleaning and normalization, data from 592 depth points make up the training and test dataset. Four hundred and seventy-five depths were randomly selected as the training data, and the remaining 100 depths were used as the testing data.

4 VAEc-based NN model

4.1 Training efficiency

The VAEc model constructed in Sect. 2.3 is trained on a desktop computer with 3.5 GHz CPU and 32 GB RAM. It takes 41.25 s for the two-stage training process to train the model 1000 times with training data, including 10 logs and

64-bin T2 distribution, from 460 discrete depths. Training loss and prediction accuracy of the model are monitored to select appropriate number of steps for the training. The training steps are selected by monitoring the mean square error (MSE) of the learning on both training and testing datasets. In the initial training stage, MSEs for training and testing decrease as the model learns from the training dataset. After certain number of steps of training, the testing MSE tends to increase, and this is the point where the model starts overfitting the datasets. We stop training at this training step to avoid overfitting. We set the training steps to 2000, for which the model has a low loss and high prediction accuracy. In the first step of training, VAEc generalizes the NMR T2 in the training dataset and the decoder generates a typical NMR T2 from the 3-dimensional latent vector. Figure 3 is a 2D sample from the 3-dimensional latent space generated by the encoder. The figure represents the learning achieved by the VAEc-based NN model.

Figure 3 is a slice plane from the 3D latent space. It demonstrates how the latent vectors are related to the NMR T2 distributions. The VAEc learns from the training dataset and generalizes the typical NMR T2 shape in the latent space. The decoder can decode a latent vector into an NMR T2 distribution. Each NMR T2 in Fig. 3 corresponds to a 3-dimensional vector, which represents a coordinate in the space. By sampling from the latent space, we can visualize VAEc's understanding of a typical NMR T2 distribution. In Fig. 3, the learned T2 distribution changes gradually in the 3D latent space with slight consecutive variations in the number of peaks, smoothness of the distribution, the position of the peaks, and height and width of the peak. The VAEc learning ensures that the gradual changing characteristic of the learned NMR T2 plotted in Fig. 3 is in accordance with the characteristic of NMR T2 in the formation. NMR T2 in adjacent formations should be similar to each other with gradual changing characteristics. After the first stage of training, the second stage relates the learned features in NMR data to the input logs used for the synthesis.

4.2 Testing process and synthesis accuracy

The synthesis accuracy of the trained VAEc-based NN is tested using the 100 testing data. R^2 values are calculated using the predicted NMR T2 and measured NMR T2 distributions, such that

$$R^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (2)$$

where $\hat{y}_i$ and y_i are predicted and actual NMR T2 amplitude at bin i , and n is the number of bins.

Fig. 3 Examples of the learned manifold from the training dataset. *x*-axis represents T2 relaxation time from 0.3 to 3000 ms; *y*-axis is normalized NMR signal amplitude approximately ranging from 0 to 0.25

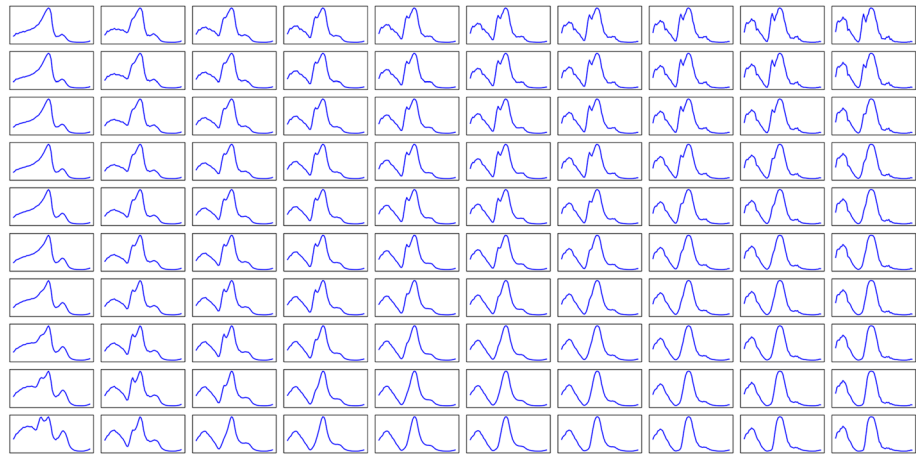
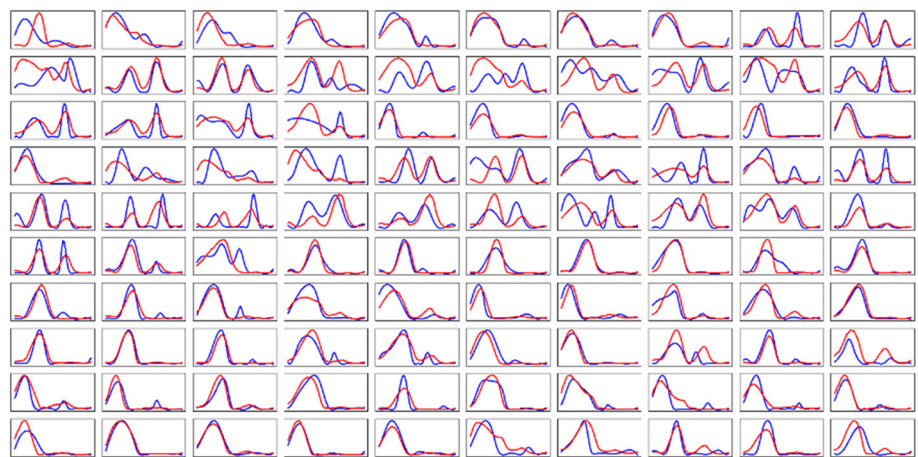


Fig. 4 True NMR T2 (blue) and predicted NMR T2 (red) in the testing phase. *x*-axis represents T2 relaxation time from 0.3 to 3000 ms; *y*-axis is normalized NMR signal amplitude approximately ranging from 0 to 0.25 (color figure online)



The results are shown in Fig. 4. The average R^2 value for the 100 predictions of T2 distributions in test data is 0.75. In Fig. 4, the randomly selected NMR T2 testing data show a variety of shapes. About 60% of 100 samples are single-peak NMR T2 and 40% are double-peak. For single-peak NMR T2, the predicted NMR T2 and true NMR T2 almost overlay with each other. For double-peak NMR T2, the accuracy is lower than single-peak data. Although the VAEc model can predict the peak position of double-peak NMR T2 with high accuracy, the model has low prediction accuracy for the height and shape of peaks.

Distribution of R^2 of prediction for the 100 testing data is shown in Fig. 5. R^2 achieved for most depths is above 0.7. However, VAEc model has lower synthesis accuracy at depths exhibiting 2-peaked T2 distribution. This is primarily due to the limited training data with 2-peaked NMR T2 distribution. In the 300-ft thick Bakken shale formation studied in this work, close to 1/3 of the depths have 2-peaked T2 distribution. Variation in the shape of T2 distribution indicates changes in formation characteristics and changes in the relationships between T2 and formation mineral content and fluid saturation. In the absence of large

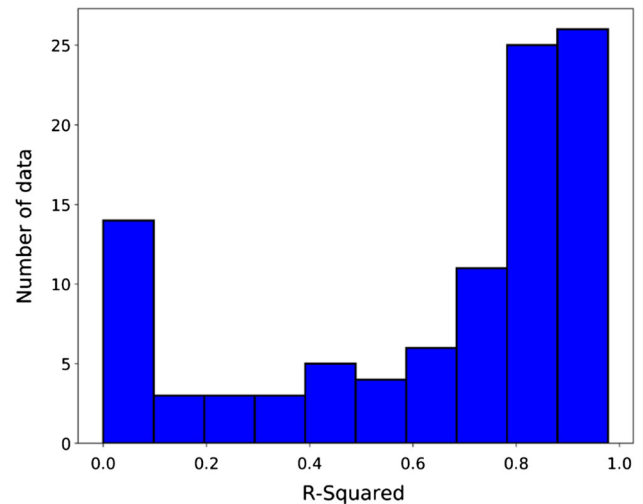


Fig. 5 R^2 distribution representing synthesis accuracy of the VAEc model during the testing phase

volume of suitable training data, the proposed data-driven predictive models will not be able to accurately determine the mathematical relationships between T2 distributions

and the input logs, thereby leading to poor synthesis accuracy for depths having physical characteristics that are not adequately represented in the training dataset.

4.3 Sensitivity to noise in input logs

Environmental noise and uncertainty due to physical variability affect well logs. We studied these adverse effects on the synthesis accuracy by adding 50% noise to each of the 10 input logs, one at a time, and then studying the corresponding variations in synthesis accuracy across the entire T2 distribution spectrum. Data preprocessing is performed to normalize each of the 10 input logs between -1 and 1 . Noise data are generated from a uniform distribution ranging between -0.5 and 0.5 and added to the normalized data resulting in addition of 50% noise to the input logs. VAEc model is trained 10 times, wherein 50% noise is added to one of the 10 logs, one at a time. During each time of training, the VAEc model is trained for 2000 training steps, during which the weights inside the network are updated by backpropagation of the loss. Our numerical experiments indicate overfitting when the training steps are higher than 2000.

Mean square error (MSE) in the synthesis of fluid-filled pore size distribution, approximated as T2 distribution, using noise-prone logs and original logs is calculated for this sensitivity study. MSE is defined as

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (3)$$

where $\hat{y}_i$ and y_i are predicted and actual NMR T2 amplitude at bin i and n is the number of bins. MSE variation when performing the synthesis using the noise-prone logs with respect to the MSE of synthesis using the original logs is shown in Fig. 6. Positive MSE variation indicates that the

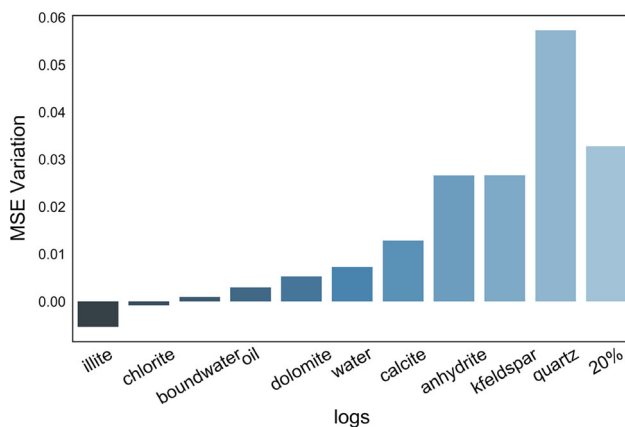


Fig. 6 Variation in MSE of the proposed synthesis when 50% noise is added to each of the 10 logs (one at a time). The bar on the right is the MSE variation when 20% noise is added simultaneously to each input log

addition of noise to logs lowers the synthesis accuracy. When 50% noise is added to the quartz and illite content logs, MSE increases by 5.72% and decreases by 0.54%, respectively. The robustness of the proposed methodology is demonstrated with this sensitivity study because the change in MSE due to noise is minor despite adding a large amount of noise to one log. Figure 6 indicates that the synthesis accuracy is sensitive to noise in quartz, feldspar, anhydrite, and calcite content logs, whereas the synthesis accuracy is insensitive to noise in oil saturation, bound water saturation, chlorite content, and illite content logs.

To evaluate the importance of input logs in the proposed synthesis, we study the change in synthesis accuracy when one of the 10 input logs is not used in the synthesis methodology. The model is trained 10 times; in each training, one of the 10 logs is removed. The model is trained for 2000 steps during each training with the 9 remaining input logs. MSE variation due to the removal of one log is shown in Fig. 7. The proposed synthesis methodology is most affected when bound water saturation and illite content logs are removed. MSE increases about 11% when bound water and illite logs are removed. Quartz, calcite, and feldspar content logs have intermediate adverse effects on the synthesis accuracy. In contrast, water saturation, anhydrite content, dolomite content, and oil saturation logs when removed result in slight improvement in the synthesis accuracy.

4.4 Sensitivity to noise in NMR T2

Noise contained in NMR T2 distributions used during the training phase will introduce inaccuracy in the synthesis. Simulating the effect of noise in T2 distribution requires a different approach compared to that used for studying the effect of noise in input logs. To add noise into T2 distribution while keeping its original characteristic, we first applied Gaussian fitting to fit each NMR T2 with 3

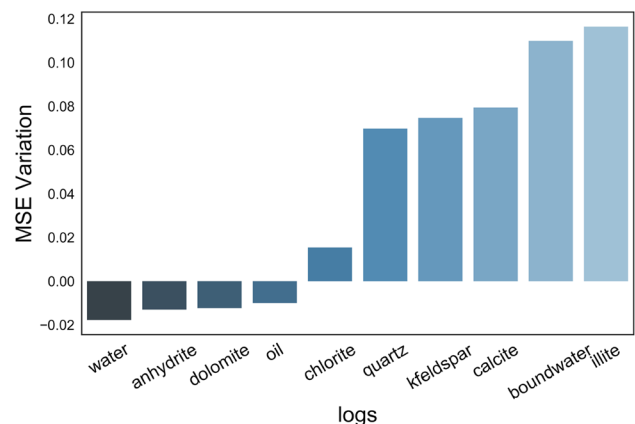


Fig. 7 MSE variation when one input log is removed one at a time

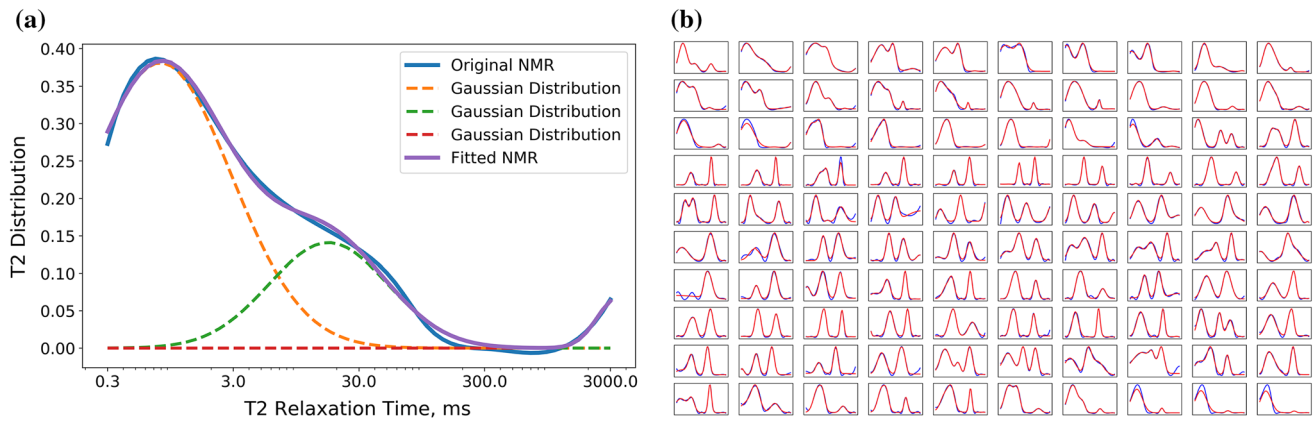


Fig. 8 **a** Demonstration of fitting NMR T2 with 3 Gaussian distributions, and **b** examples of multi-Gaussian fits (red) to the original (blue) NMR T2 without any additional noise (color figure online)

Gaussian distributions as shown in Fig. 8a. Each Gaussian distribution is controlled by 3 parameters of position, amplitude, and width. In this way, each NMR T2 is described by 9 parameters. By slightly changing the 9 controlling parameters, we can change the peak amplitude, position, and width of the measured T2 distribution used for the training. We added 30% of random noise to each of the 9 parameters to add noise to the measured NMR T2 while keeping its original characteristic. The multi-Gaussian fits to the measured T2 distributions are shown in Fig. 8b. The fit is near perfect. The noise-prone T2 distributions generated using the aforementioned approach are shown in Fig. 9. By adding noise to the controlling parameters, the noisy T2 keeps the overall characteristics and the median R^2 value between the original NMR T2 and noisy NMR T2 is 0.69.

The VAEc model is trained with 10 original logs without additional noise and NMR T2 containing the synthetically added 30% noise to the 9 controlling parameters. The

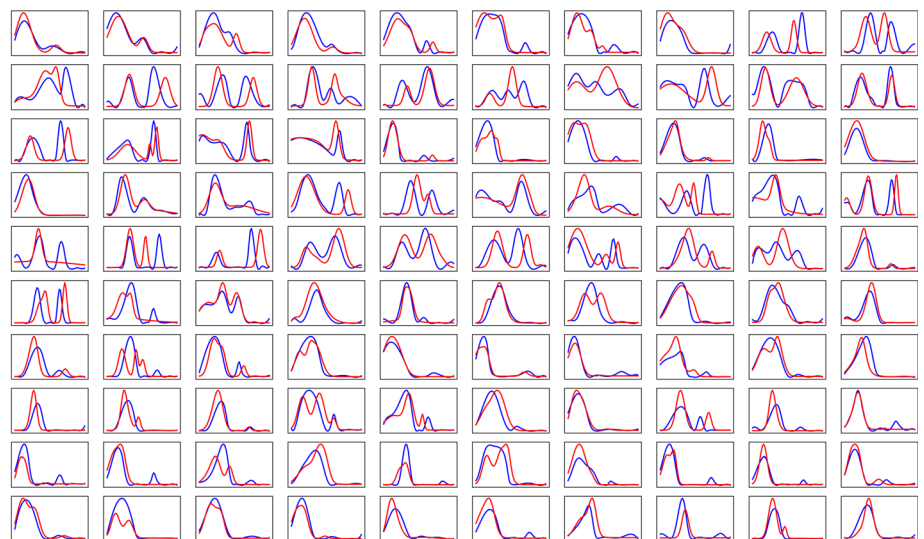
architecture and training steps remain the same. The synthesis accuracy of VAEc model decreases slightly due to the noise in the T2 data used for the training. Synthesis of T2 distribution with noise in the training T2 data has an R^2 of 0.72, which for training with noise-free original NMR T2 was 0.75. This demonstrates the robustness of the proposed synthesis methodology to noise in the training T2 data.

5 LSTM model

5.1 Training efficiency

LSTM model is trained and tested with the same dataset as the one used for the VAEc-based NN model, described in the above section. LSTM requires more time to train. It took 566.33 s to train the LSTM model with 1000 training steps, whereas it took only 41.25 s for the VAEc-based NN

Fig. 9 Original NMR T2 (blue) without additional noise and NMR T2 with synthetically added 30% noise (red). x-axis represents T2 relaxation time from 0.3 to 3000 ms; y-axis is normalized NMR signal amplitude approximately ranging from 0 to 0.25 (color figure online)



model training with the same steps. The two neural networks use different training processes. The VAEc-based model has 1825 parameters to train, whereas the LSTM model has 2071 parameters. Even though the two models have a similar number of parameters, the differences in calculation process cause one-order difference in the training times. The LSTM model takes one of the 10 logs at each time step. Besides, when predicting the NMR T2 distribution, the LSTM model predicts one value of the 64-dimensional NMR T2 at each time step, which means it takes 64 time steps to predict all values of the NMR T2. This calculation process is more complicated than the VAEc-based NN model.

5.2 Testing and accuracy

The average synthesis accuracy of the LSTM model in terms of R^2 is 0.78. The predicted T2 distributions and the R^2 of the synthesis performance are shown in Figs. 10 and 11. LSTM model better predicts single-peak T2 distribution compared to those with double peaks. In Fig. 11, the synthesis performance exhibits higher frequency of occurrence above R^2 of 0.9 or below 0.1, as compared to the performance of VAEc-based NN model, shown in Fig. 5. We claim that the LSTM model predicts single-peak T2 more accurately than the VAEc-based model, whereas the LSTM performance in predicting the double-peak T2 distribution is worse than that of VAEc-based NN model. Several LSTM synthesis for one-peak NMR do not demonstrate the typical T2 shape, for example, the 5th T2 distribution in the last row and the 6th distribution in the fourth row.

Compared to the VAEc-based NN model, the LSTM model has more freedom making it easier to overfit. First, the decoder part of the VAEc model is pretrained, which constrains the capacity of the model. Second, the LSTM model has more parameters compared with the VAEc

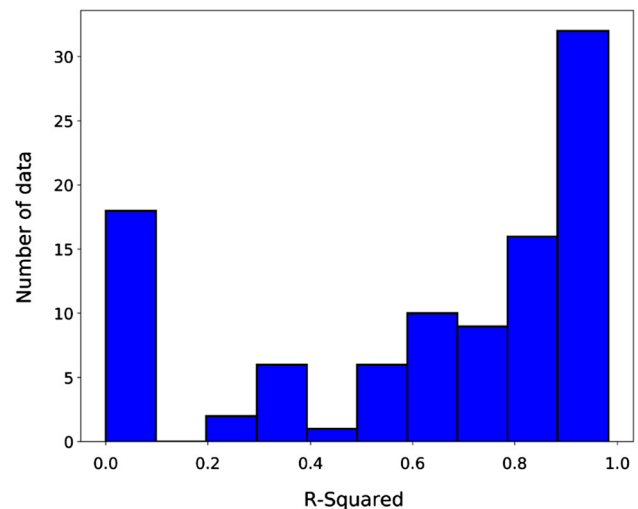


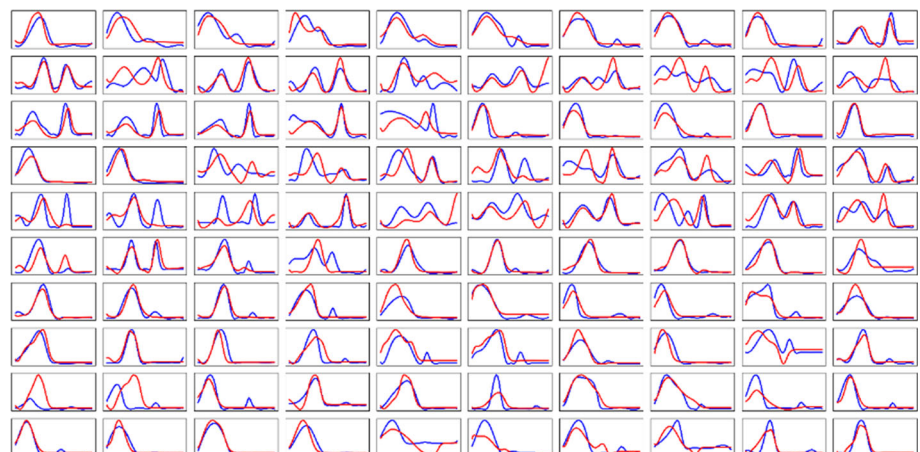
Fig. 11 R^2 distribution representing synthesis accuracy of the LSTM model during the testing phase

model. The decoder trained in the first stage of VAEc training when used in the second stage of the training constrains the capacity of the VAEc-based NN model. The VAEc architecture forces reconstruction of the NMR T2 distribution through the decoder. The freedom of the LSTM model may lead to overfitting to some extent, and overfitting leads to more dispersed synthesis accuracy in the entire formation. LSTM predicts more accurately with more freedom, but it also overfits samples.

5.3 Sensitivity to noise analysis

The model sensitivity is also tested for LSTM model using same procedure demonstrated in Sect. 4.3, by adding 50% noise to each log. Synthetically added noise in water saturation, chlorite content, k-feldspar content, and anhydrite content reduces the synthesis accuracy in comparison with that achieved using LSTM to process input logs without additional noise. Notably, the addition of noise to dolomite

Fig. 10 True NMR T2 (blue) and predicted NMR T2 (red) in testing phase. x-axis represents T2 relaxation time from 0.3 to 3000 ms; y-axis is normalized NMR signal amplitude approximately ranging from 0 to 0.25 (color figure online)



content log and bound water-saturation log improves the synthesis accuracy. This result is different with that of the VAEc model. In VAEc model, the noise in quartz content, k-feldspar content, and anhydrite content drastically decreases the synthesis accuracy. The difference of the MSE variation between VAEc and LSTM model indicate that the two models relies on different hidden relationship to predict NMR T2 from formation composition logs. Both LSTM and VAEc-based NN models are adversely affected when k-feldspar content, anhydrite content, calcite content, and oils saturation logs contain noise. VAEc model is more sensitive to noise in logs compared to the LSTM model, especially for the quartz content log (Fig. 12).

Changes in synthesis accuracy when 1 of the 10 input logs is removed from the LSTM model is studied in this section. Overall, the chlorite, anhydrite, and illite content logs when individually removed have the largest adverse effect on the synthesis. The bound water saturation and dolomite content logs when individually removed result in the improvement of the synthesis accuracy. For LSTM and VAEc-based NN models, the effects of removal of one of the 10 logs are completely opposite, except for the dolomite content log, which when removed results in a small improvement (Fig. 13).

5.4 Sensitivity to noise in NMR T2

Similar to the procedure in Sect. 4.4, we studied the effect of noise in the measured T2 data used for training on the synthesis accuracy. Contrary to the VAEc model, which shows a good stability to noise in the training T2 data, the synthesis accuracy of LSTM model significantly deteriorates on adding noise to the T2 data. Synthesis accuracy quantified in terms of R^2 drops from 0.78 to 0.43 when the training T2 data contain 30% noise. The drop in the

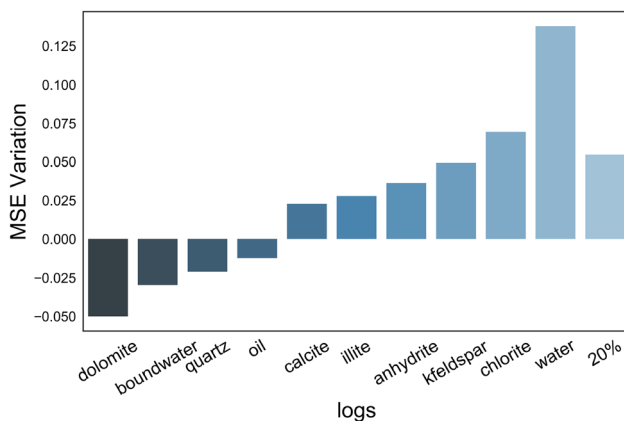


Fig. 12 Variation in MSE of the proposed synthesis when 50% noise is added to each of the 10 logs (one at a time). The bar on the right is the MSE variation when 20% noise is simultaneously added to each of the 10 input logs

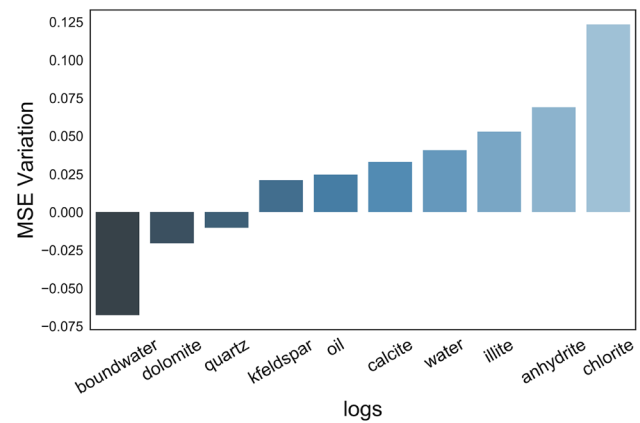


Fig. 13 MSE variation when one input log is removed one at a time

prediction accuracy indicates that the LSTM architecture is sensitive to noise in NMR T2. Unlike the VAEc model, the LSTM model takes NMR T2 as sequences. The LSTM model has the ability to find long- and short-term dependency in input and in output data. The prediction of NMR T2 values at higher relaxivity time is affected by the NMR T2 values at lower relaxivity time when using the LSTM model. Adding noise in the NMR T2 changes the dependency among the 64 bins of the NMR T2 distribution; thus, the LSTM model fails to capture the dependencies. VAEc-based NN model has a pretrained decoder, which ensures the most dominant features in the noisy NMR distributions are captured for purposes of synthesis during the testing phase.

5.5 Originality of the proposed research methods and findings

- Downhole nuclear magnetic resonance (NMR) logs acquired in the borehole environment are valuable for subsurface characterization because they contain information about the pore size distribution, fluid composition, fluid saturation, fluid mobility, formation permeability, and porosity. NMR log acquisition can be challenging due to operational and financial constraints. NMR T2 distributions of the subsurface earth formations when generated by processing conventional well logs using deep learning neural network (NN) models improve the accessibility to subsurface pore size distributions, residual saturations, movable fluid content, and permeability.
- Compared to other conventional well logs, NMR log acquisition encounters financial and operational constraints during the subsurface deployment of the NMR tool in the borehole environment. Consequently, NMR tool is deployed in very few wells, whereas conventional logging tools are run in almost

all the wells. To overcome these constraints, NMR T2 distribution of the subsurface earth is generated by processing easy-to-acquire conventional logs using deep learning neural network (NN) models.

- (c) Recent advances in machine learning and deep learning enable extraction of complex hidden relationships in a dataset and perform predictions based on the learned relationships. The ability of conventional multilayer neural network to find relationships in high-dimensional data is limited. Our prediction problem involves establishing relationships between 10 scalar measurements and 64-dimensional spectral response. To solve such a high-dimensional prediction problem, we implemented deep neural networks with specially designed architecture that honors the physical phenomena influencing the data.
- (d) For the first time, 10 conventional logs are integrated for the desired NMR T2 generation which is very different from the typical CNN and LSTM applications on images and audio.
- (e) CNN honors the image-like spectral behavior of the T2 distribution at a given depth that requires multi-scale filters for efficient extraction of features, whereas LSTM honors the time series-like short-term and long-term dependencies of conventional log inputs and T2 distributions along the depth of a geological formation. There has not been any research work on using these techniques for improving subsurface characterization under data constraints.
- (f) A comparative study of VAEc versus LSTM capabilities in generation of spectral response of geomaterial sheds light on the limitations and advantages of the two distinct architectures, one meant for image processing and the other meant for audio processing. We found that when there is noise in T2 distribution used for training, VAEc model has stable performance, whereas the R^2 of LSTM model drops from 0.78 to 0.43.

5.6 Limitations and challenges of proposed research methods and findings

- (a) In this pioneering work, the data being used for the training/testing are limited in quality and quantity due to proprietary concerns of the oil and gas company that donated the data. Lack of NMR data is primarily due to the high cost of the NMR logs and the proprietary nature of this data that is not easily shared by the oil and gas companies. Access to a larger dataset is required for validating the prediction performance of the proposed methodology.

- (b) The 10 conventional logs and NMR T2 distribution logs used for training the neural network model were recorded during different logging runs performed in a 5-inch-diameter borehole intersecting the geological formation under investigation. Consequently, the training dataset is prone to noise and adversely affected by the heterogeneity, complexity, and uncertainty due to the surrounding borehole and near-wellbore environments. This will affect the prediction performance of the deep learning neural networks for the proposed application.

We test the methodology on well logs from a single well. Cross-well validation is needed to test the robustness of the methodology. Due to lack availability of NMR logs and their proprietary nature, cross-well validation has not been performed in this study.

6 Summary of results

A neural network model based on variational autoencoder with convolutional layers (VAEc-based NN model) and a ‘long short-term memory’ recurrent neural network model (LSTM model) are trained and tested to synthesize in situ fluid-filled pore size distribution in the Bakken shale formation. To that end, the two models process 7 formation mineral content and 3 fluid saturation logs. The training/testing data were measured in an open borehole environment across a 300-ft depth interval of the Bakken shale formation. LSTM and VAEc-based NN models have a similar number of parameters and are trained/tested on the same dataset; however, LSTM incurs one order of magnitude higher training time. The synthesis accuracy for VAEc-based NN and LSTM models is 0.75 and 0.78, respectively. VAEc-based NN model can synthesize NMR T2 distributions, which approximates the in situ fluid-filled pore size distribution, in the presence of 50% noise in input logs and 30% noise in training T2 data without significant deterioration in the synthesis accuracy. On the other hand, LSTM model is significantly affected by noise in the training T2 data and robust in the presence of 50% noise in the input logs. Synthesis accuracy of LSTM model quantified in terms of R^2 drops from 0.78 to 0.43 when the training T2 data contain 30% noise. LSTM model is more sensitive to noise in chlorite content, k-feldspar content, and water saturation logs, whereas VAEc model is more sensitive to noise in quartz, anhydrite, and k-feldspar content logs. The information contained in 7 formation mineral content and 3 fluid saturation logs has redundancy, such that removing one of the 10 logs does not significantly affect the synthesis accuracy. LSTM and VAEc-based NN models exhibit completely opposite sensitivity to removal

of anyone of the 10 input logs, indicating that the two models extract different hidden log relationships for purposes of generating NMR T2 distributions, which approximates the in situ fluid-filled pore size distribution.

Acknowledgements Few aspects of the recurrent neural network model implementation are based upon the work supported by the Geosciences Research Program in the Office of Basic Energy Sciences, U.S. Department of Energy, under Award Number DE-SC0019266. We thank Mr. Gary Simpson, Dr. Carl Sondergeld, and Dr. Chandra Rai for their technical insights and suggestions related to nuclear magnetic resonance measurements and other well logs.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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